
EXPERIENCE**Omnicom Adobe Practice**

Remote

*Backend Developer**Jan 2023 – Present*

- Optimized content management systems for Fortune 500 corporations by applying industry best practices
- Automated the propagation of metadata across a 150+ terabyte library, eliminating manual updates
- Developed a configurable scheduler that queries and triggers workflows to automate the archival of assets
- Provided users with an interface to manage storyboards by using asynchronous JavaScript for CRUD operations
- Automated user group access control for new asset collections by programmatically enforcing policy rules

Altra Interactive

Remote

*Software Developer**Jan 2022 – Dec 2023*

- Maintained internal tools for accelerating client-work by fixing issues, fielding questions, and managing contributors
- Designed a sample project based on common feature requests to prepare new developers for client work
- Accelerated data retrieval for content authors by making an application that converts form data into query strings
- Introduced new and relevant web development technologies, sharing insights and training team members
- Analyzed client requirements in discovery sessions, defining project scope, user stories, and actionable tasks

Michigan Institute for Data Science

Ann Arbor, MI

*Research Assistant**May 2021 – Jan 2022*

- Developed a full stack application for a Native American education department to share opportunities with students
- Secured \$50,000 in funding, establishing a reusable model to promote data sovereignty for underserved communities
- Designed a dashboard to view student data and analyze learning outcomes by collaborating with administrators
- Prioritized privacy and security by implementing a strong password strategy that utilized cryptographic hashing
- Consolidated, validated, and migrated student data from various sources to a cloud-hosted document store
- Managed and documented process of deploying to a cloud application platform to improve maintainability

PROJECTS**Vid2Gif**github.com/stavguo/vid2gif

- Created a desktop app that converts videos to GIFs by managing FFmpeg and Gifski sidecar processes in Rust
- Improved usability by coupling FFmpeg to decode various video formats and piping frames into the Gifski encoder
- Provided video cropping and frame selection through a responsive frontend using SvelteKit and TailwindCSS

Pathfinding Solution with Respect to Movement Costsgithub.com/stavguo/pico-saga

- Architected a lightweight game engine, fitting pathfinding, AI, and rendering systems into a 64kb memory footprint
- Implemented a custom pathfinding solution, enabling agent navigation across procedurally generated terrain
- Designed a finite state machine to manage complex entity behaviors and game-state logic to improve code clarity

Synthesizing Realistic Urban Road Networksgithub.com/stavguo/obb-parcelling

- Increased realism in simulating urban environments by creating an interactive program to subdivide city blocks
- Recursively split polygons based on their minimum bounding box to concisely solve the block subdivision problem

Markdown-Driven Content Management Systemgustavo.zip

- Developed a portfolio website that leverages Go's templating language to convert Markdown to rendered HTML
- Documented research projects, personal endeavors, interview insights, and other topics to track personal growth

SKILLS**Languages:** C, C++, C#, CSS, Go, HTML, Java, JavaScript, Python, SQL, TypeScript**Frameworks & Tools:** AEM, Azure DevOps, Docker, Express.js, Git, MongoDB, Node.js, PostgreSQL, React, REST APIs**Cloud & Deployment:** AWS Lambda, Azure, GCP**Libraries & Testing:** Axios, Bootstrap, Jest, JUnit, Mockito, Redux

EDUCATION**University of Michigan**

Ann Arbor, MI

*Bachelor of Science in Computer Science**Sept 2018 – Dec 2021*